

30 CHAPTERWISE IMPORTANT QUESTIONS

Mathematics Standard (041) | CBSE 2026-27

Surface Areas and Volumes

CBSE Mathematics 2026-27 | Board-examinable

20 numerical / problem-solving questions and 10 theory, proof or method questions. Workings and answers are written for Class X revision.

Question 1

Find the volume of a cylinder of radius 3 cm and height 8 cm, in terms of π .

Answer:

$$\text{Volume} = \pi r^2 h = \pi \times 3^2 \times 8 = 72\pi \text{ cm}^3.$$

Question 2

Find the volume of a cylinder of radius 4 cm and height 10 cm, in terms of π .

Answer:

$$\text{Volume} = \pi r^2 h = \pi \times 4^2 \times 10 = 160\pi \text{ cm}^3.$$

Question 3

Find the volume of a cylinder of radius 5 cm and height 12 cm, in terms of π .

Answer:

$$\text{Volume} = \pi r^2 h = \pi \times 5^2 \times 12 = 300\pi \text{ cm}^3.$$

Question 4

Find the volume of a cylinder of radius 7 cm and height 14 cm, in terms of π .

Answer:

$$\text{Volume} = \pi r^2 h = \pi \times 7^2 \times 14 = 686\pi \text{ cm}^3.$$

Question 5

Find the volume of a cylinder of radius 2 cm and height 9 cm, in terms of π .

Answer:

$$\text{Volume} = \pi r^2 h = \pi \times 2^2 \times 9 = 36\pi \text{ cm}^3.$$

Question 6

Find the volume of a cylinder of radius 6 cm and height 15 cm, in terms of π .

Answer:

$$\text{Volume} = \pi r^2 h = \pi \times 6^2 \times 15 = 540\pi \text{ cm}^3.$$

Question 7

Find the volume of a cylinder of radius 10 cm and height 20 cm, in terms of π .

Answer:

$$\text{Volume} = \pi r^2 h = \pi \times 10^2 \times 20 = 2000\pi \text{ cm}^3.$$

Question 8

Find the volume of a cylinder of radius 3 cm and height 12 cm, in terms of π .

Answer:

$$\text{Volume} = \pi r^2 h = \pi \times 3^2 \times 12 = 108\pi \text{ cm}^3.$$

Question 9

Find the volume of a cylinder of radius 4 cm and height 7 cm, in terms of π .

Answer:

$$\text{Volume} = \pi r^2 h = \pi \times 4^2 \times 7 = 112\pi \text{ cm}^3.$$

Surface Areas and Volumes - Continued

Question 10

Find the volume of a cylinder of radius 5 cm and height 9 cm, in terms of pi.

Answer:

$$\text{Volume} = \pi r^2 h = \pi \times 5^2 \times 9 = 225\pi \text{ cm}^3.$$

Question 11

Find the curved surface area of a cone with radius 3 cm and height 4 cm, in terms of pi.

Answer:

$$\text{Slant height } l = \sqrt{3^2 + 4^2} = 5. \text{ CSA} = \pi r l = 15\pi \text{ cm}^2.$$

Question 12

Find the curved surface area of a cone with radius 5 cm and height 12 cm, in terms of pi.

Answer:

$$\text{Slant height } l = \sqrt{5^2 + 12^2} = 13. \text{ CSA} = \pi r l = 65\pi \text{ cm}^2.$$

Question 13

Find the curved surface area of a cone with radius 6 cm and height 8 cm, in terms of pi.

Answer:

$$\text{Slant height } l = \sqrt{6^2 + 8^2} = 10. \text{ CSA} = \pi r l = 60\pi \text{ cm}^2.$$

Question 14

Find the curved surface area of a cone with radius 7 cm and height 24 cm, in terms of pi.

Answer:

$$\text{Slant height } l = \sqrt{7^2 + 24^2} = 25. \text{ CSA} = \pi r l = 175\pi \text{ cm}^2.$$

Question 15

Find the curved surface area of a cone with radius 8 cm and height 15 cm, in terms of pi.

Answer:

$$\text{Slant height } l = \sqrt{8^2 + 15^2} = 17. \text{ CSA} = \pi r l = 136\pi \text{ cm}^2.$$

Question 16

Find the curved surface area of a cone with radius 9 cm and height 12 cm, in terms of pi.

Answer:

$$\text{Slant height } l = \sqrt{9^2 + 12^2} = 15. \text{ CSA} = \pi r l = 135\pi \text{ cm}^2.$$

Question 17

Find the curved surface area of a cone with radius 5 cm and height 5 cm, in terms of pi.

Answer:

$$\text{Slant height } l = \sqrt{5^2 + 5^2} = 7.07107. \text{ CSA} = \pi r l = 35.3553\pi \text{ cm}^2.$$

Question 18

Find the curved surface area of a cone with radius 12 cm and height 16 cm, in terms of pi.

Answer:

$$\text{Slant height } l = \sqrt{12^2 + 16^2} = 20. \text{ CSA} = \pi r l = 240\pi \text{ cm}^2.$$

Question 19

Find the curved surface area of a cone with radius 4 cm and height 3 cm, in terms of pi.

Answer:

$$\text{Slant height } l = \sqrt{4^2 + 3^2} = 5. \text{ CSA} = \pi r l = 20\pi \text{ cm}^2.$$

Surface Areas and Volumes - Continued

Question 20

Find the curved surface area of a cone with radius 10 cm and height 24 cm, in terms of pi.

Answer:

Slant height $l = \sqrt{10^2 + 24^2} = 26$. CSA = $\pi r l = 260\pi \text{ cm}^2$.

Question 21

State the main formula, rule or theorem used in Surface Areas and Volumes.

Answer:

Volume of cylinder = $\pi r^2 h$; volume of cone = $\frac{1}{3} \pi r^2 h$; volume of sphere = $\frac{4}{3} \pi r^3$.

Question 22

What do the important symbols or quantities in Surface Areas and Volumes represent?

Answer:

r is radius, h is height and l is slant height.

Question 23

Write the correct method for solving a Surface Areas and Volumes numerical.

Answer:

Draw the solid, decide whether curved or total area is asked, calculate any missing slant height, then substitute.

Question 24

Which unit, sign convention or condition must be checked in a Surface Areas and Volumes answer?

Answer:

Write the given data first, use the stated convention, and include units wherever the final quantity has a unit.

Question 25

Why must the final answer in Surface Areas and Volumes be checked against the given condition?

Answer:

Checking catches wrong signs, arithmetic errors and values that are impossible in the original situation.

Question 26

What is one common error in Surface Areas and Volumes questions?

Answer:

Students often substitute a value in the wrong formula or skip a condition. Write the formula before substituting.

Question 27

How should full working be shown in a Surface Areas and Volumes numerical?

Answer:

Show the formula or theorem, substitution, simplification, calculation and final answer in that order.

Question 28

When is a diagram, table or graph useful in Surface Areas and Volumes?

Answer:

Use it when it makes the given quantities and their relationship clear before calculation.

Surface Areas and Volumes - Continued

Question 29

How can a student decide which formula to use in Surface Areas and Volumes?

Answer:

Identify the quantity asked, list the known quantities, then select the formula or theorem connecting them.

Question 30

What makes a high-scoring answer in Surface Areas and Volumes?

Answer:

Correct reasoning, complete steps, an appropriate formula, a clearly stated final result and a quick verification.

Sources: [CBSE Mathematics curriculum 2026-27](#); [CBSE Class X sample papers and marking schemes 2026-27](#); [NCERT textbooks](#).

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